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# DIGRD: Disguised Inertial Gait Recognition Dataset and Method

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***Abstract*—Inertial gait recognition, renowned for its contactless operation and long-range identification capabilities, enables user-friendly and secure biometric authentication. However, the systematic evaluation of inertial gait recognition under clothing and footwear disguises, a critical factor in real-world security applications, has not been addressed. This paper introduces DIGRD (Disguised Inertial Gait Recognition Dataset), the first benchmark dataset explicitly designed to study disguised inertial gait recognition. Constructed in a controlled indoor building, DIGRD contains 61,346 gait events from 40 subjects across 4 clothing-footwear covariates. Experimental results demonstrate that both clothing and footwear changes significantly degrade recognition performance, with footwear alterations exerting a stronger deceptive effect (84.85% vs. 88.78%). This work establishes a benchmark for studying inertial gait recognition under disguise conditions and paves the way for developing more secure biometric authentication systems.**

***Keywords—gait recognition, inertial data, disguise covariates***

## I. Introduction

Gait recognition, a biometric technology leveraging unique human locomotion patterns, has gained significant momentum in recent years, due to its non-contact, long-range, and privacy-preserving advantages [1]. Advancements in computer vision and pattern recognition have enabled the development of diverse gait datasets. For instance, the SUSTech1K dataset that focuses on 3D point cloud analysis [2], the CCPG dataset that deeply explores gait recognition under cloth-changing conditions [3], and the CCGR dataset that proposes a cross-covariate gait recognition benchmark [4]. Nevertheless, these datasets primarily focus on visual or silhouette-based methods and do not consider how clothing and footwear changes alter the body's center of gravity, nor how inertial sensors perceive these subtle but significant biomechanical shifts in gait dynamics.

While deep learning has propelled gait recognition performance, practical deployments remain challenged by viewpoint variations, environmental noise, and deliberate deception tactics [5]. Existing visual datasets (CASIA-B [6], OU-ISIR [7]) partially address clothing effects but rely on silhouette-based features, making them inherently vulnerable to visual obstruction. To address this gap, our previous work introduced the AFPID-II dataset [8] for acoustic footstep recognition and the DisGait dataset [9] targeting visual disguises and posture variations. Results showed that acoustic methods are sensitive to environmental noise, while visual disguises decrease silhouette-based recognition performance by about 40%. However, few public datasets investigate inertial-based gait recognition under disguise conditions.

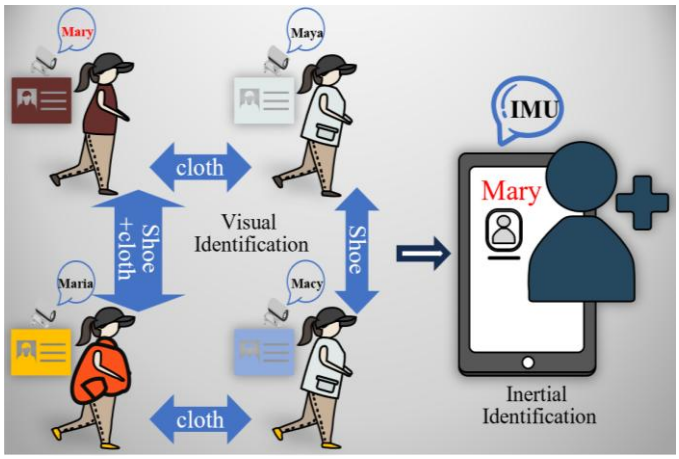


Fig. 1. Illustration of visual vs. inertial gait recognition under disguises. Visual identification may fail when external appearance changes, while inertial recognition, based on motion dynamics, is less affected. The figure highlights the potential robustness of inertial methods in disguised scenarios.

Based on existing research and our prior work, current gait recognition datasets have explored sound and visual cues to some extent. However, recognition accuracy under disguised conditions remains low due to interference. Notably, few public datasets specifically target inertial-based gait recognition under disguise. To address this limitation, we constructed the DIGRD dataset, the first publicly available inertial gait dataset designed to investigate how clothing and footwear affect recognition by altering the body's center of gravity. DIGRD includes two covariates: clothing (normal attire (N), thick coat (C), lab coat (W)) and footwear (two types (S)). As shown in Fig. 2, participants performed gait recordings under varying conditions, resulting in 61,346 gait events from 40 university students without movement impairments.

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We also carry out fundamental experiments, employing both traditional machine learning and deep learning approaches to inertial gait recognition. The main contributions of this paper are as follows: (I) Systematically collected the first disguised inertial gait recognition dataset incorporating clothing and footwear. (II) Developed a deep neural network-based method for disguised inertial gait recognition. (III) Conducted experiments to evaluate the impact of clothing and footwear changes on gait recognition. The DIGRD dataset and corresponding baseline implementation will be made publicly available shortly at the GitHub Repository[1] to facilitate reproducibility and further research.

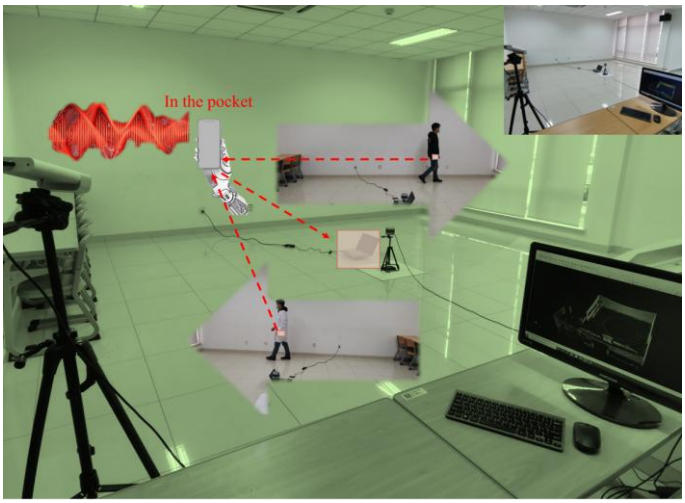


Fig. 2. A snapshot of data recording. Participants were instructed to walk naturally around the data collection setup while carrying a smartphone equipped with inertial sensors (tri-axis accelerometer and gyroscope). The smartphone was uniformly placed in the participant's pants pocket to ensure consistency and realism in gait data acquisition.

## II. Related Work

### A. Disguised Gait Recognition

Gait recognition has been extensively studied across vision, audio, and inertial sensing domains, with disguise-related research focusing on clothing, footwear, and carried objects. In vision-based approaches, silhouette and skeletal features are widely used. For instance, the TUM-GAID dataset [10] defines disguise conditions as wearing a backpack or shoe covers, while SkeletonGait [11] employs skeleton maps for gait representation. These studies demonstrate that visual gait recognition is sensitive to appearance changes.

In radar-based recognition, the MMRGait dataset [12] captures gait data under varying viewing angles and attire, such as heavy coats and backpacks, further highlighting the influence of disguises. In acoustic gait recognition, our previous AFPID-II dataset [13] uses a microphone array to assess the effects of room type, clothing, and footwear. Results show that root type had the largest impact, followed by footwear changes (31%) and clothing changes (13%).

In contrast, inertial sensor-based gait recognition is less sensitive to room type and robust under disguise.

[1] https://github.com/wu-zhaoyang/DIGRD-Disguised-Inertial-Gait-Recognition-Dataset-and-Method

### B. Inertial Gait Recognition Datasets

Inertial sensors, such as accelerometers and gyroscopes, offer a non-visual, non-audio alternative for gait recognition and are widely used in mobile and wearable devices. Several datasets have been developed, including TUM-GAID [10], which involves 305 subjects and primarily explores footwear variations. The WHU Gait Dataset [14] collects data using a smartphone. In terms of methodology, DANet [15] integrates local and global motion patterns, performing well under cross-clothing and cross-footwear scenarios. GLFE [11] leverages global-local feature extraction and temporal aggregation.

Despite these advances, public datasets specifically targeting disguised inertial gait recognition remain unexplored. To address this, we propose the DIGRD dataset and introduce a deep neural network baseline.

## III. The DIGRD Dataset

The DIGRD dataset was recorded using a smartphone with inertial sensors, explicitly designed to address the challenge of clothing-footwear variations.

### A. Data Collection

We recruited 40 university students (17 females and 23 males) aged between 20 and 30, ensuring they had no mobility disorders or physical impairments.

To construct the DIGRD dataset, we considered clothing and footwear as disguise covariates. Each subject was assigned an anonymized identifier (S1-S40). And all subjects were informed about the purpose of the experiment as well as the data collection procedures. The experiment protocols were reviewed and endorsed with our former work [13].

**Theme I:** Participants wore a thick coat and the first type of footwear, walking in a counterclockwise direction.

**Theme II:** Participants removed the coat (maintaining footwear type 1) and walked in a clockwise direction.

**Theme III:** Participants changed to footwear type 2 while maintaining the coat-less condition, walking in a counterclockwise direction.

**Theme IV:** Participants wore a white lab coat over their normal clothing and walked in a counterclockwise direction.

To ensure the natural representation of disguised inertial data, participants were not given specific walking constraints, allowing them to exhibit natural arm-swinging behavior. Each data collection session lasted approximately 3.5 minutes, with breaks between sessions. Over four sessions, each participant contributed around 15 minutes (900 seconds) of walking data, resulting in approximately 1,500 individual gait events.

### B. Signal Preprocessing

Similar to Zhou et al. [14], we employed a threshold-based segmentation method to extract meaningful gait events from raw sensor data. The segmentation algorithm uses local maxima and minima of angular velocity to identify step events. A sliding window mechanism was applied to detect key gait transitions, crucial for analyzing inertial gait patterns.


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By applying this threshold-based segmentation, we extracted clean and structured inertial gait data, facilitating further gait cycle division and gait event extraction.

**Algorithm 1:** IMU-Based Gait Segmentation Algorithm

**Input:**
- $P_{\text{seg}} = \{p_i \mid i = 1, 2, \ldots, n\}$
- Thresholds for peak detection:
  - `max_threshold` = 6.0
  - `min_threshold` = 1.0
  - `peak_threshold` = 1.2
  - `threshold_roi` = 2.5

**Output:**
- Segmented gait data
- Start/end points of gait cycles
- Gait event points (local maxima/minima)

**Step 1:** Read IMU data
$p_i \leftarrow \text{readIM}(P_{\text{seg}})$ ;
**Step 2:** Noise detection
$p_{\text{raw}} \leftarrow \text{noiseDetection}(p_i)$ ;
**Step 3:** Detect local maxima
max_points $\leftarrow$ findMaxPeaks($p_{\text{raw}}$["AngVel_X"], threshold = `peak_threshold`) ;
**Step 4:** Determine termination index
end_index $\leftarrow$ findEndPoint($p_{\text{raw}}$["AngVel_X"], `max_threshold`, `min_threshold`) ;
**Step 5:** Filter data via ROI
Define ROI: $\{(x, y, z) \mid x^2 + y^2 + z^2 \leq \texttt{threshold_roi}\}$
$p_{\text{filtered}} \leftarrow \text{select}(p_{\text{raw}}, \text{ROI})$ ;
**Step 6:** Detect local minima
min_points $\leftarrow$ findMinPoints($p_{\text{raw}}$["AngVel_X"], max_points) ;
**Step 7:** Extract gait cycles
**for** $i \leftarrow 1$ **to** *length*(*max_points*) − 1 **do**
  Extract segment [max_point[$i$], max_point[$i+1$]]
  **if** *max_point*[$i+1$] > *end_index* **then**
    Extend segment to end_index
  **end**
**end**
**Step 8:** Detect gait events
**for** $i \leftarrow 1$ **to** *length*(*max_points*) **do**
  **if** *i is even* **then**
    Detect events in [max_point[$i$], min_point[$i$]]
  **else**
    **if** $i > 1$ **then**
      Detect events in [min_point[$i-1$], max_point[$i$]]
    **end**
  **end**
**end**
**Step 9:** Validate consistency
**if** *length*(*max_points*) ≠ *length*(*min_points*) **then**
  Reprocess data (return to Step 2)
**end**
**Step 10:** Return results
**return** *Segmented data, cycle boundaries, event points* ;

### C. Gait Event Extraction

Building on previous step segmentation techniques [14], we divided continuous gait data into gait cycles. A full gait cycle is defined as the time interval between two consecutive ground contacts of the same foot. Since our study focuses on single-step recognition, we considered a half-cycle for analysis. Unlike previous studies that used accelerometer data to segment gait events, our method relies on gyroscope angular velocity peaks, following the segmentation approach of Prateek et al. [16].

Our segmentation strategy utilizes local maxima and adjacent minima of angular velocity data as markers. The time interval between consecutive peak markers is identified as a complete gait cycle. By alternating between peak and valley markers, we effectively segmented step events with minimal overlap, ensuring an accurate representation of distinct gait instances. Using this segmentation approach, we compiled the DIGRD dataset by generating 61,346 gait events across four covariates.

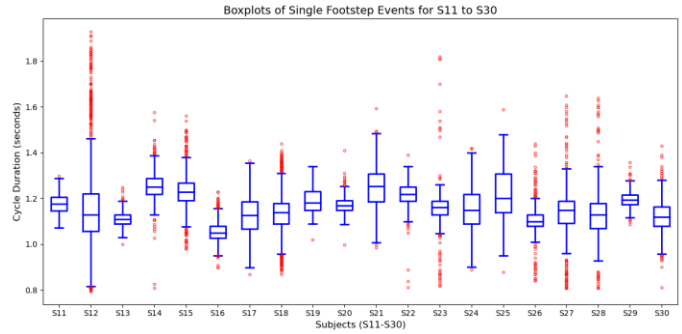


Fig. 3. Box plots of single footstep events for all 4 sessions.

### D. Gait Event Cycle Analysis

To evaluate inter-subject variability in gait dynamics, we analyzed gait cycle durations across all 40 participants. Figure 3 shows box plots for a representative subset (S11-S30). The majority fall within the range of 1.0-1.4 seconds, with closely clustered medians, indicating temporal consistency. Several subjects (S12, S16, S18, S26) exhibited outlier values, suggesting pronounced individual differences or occasional deviations in walking patterns. While most participants demonstrated narrow interquartile ranges, indicative of stable gait cycles, others (S12, S25) showed broader distributions, potentially due to idiosyncratic gait behavior or minor inconsistencies in testing conditions. Collectively, these results underscore the overall consistency and reliability of the dataset for research in disguised inertial gait recognition.

Considering that a gait cycle typically spans 1200 - 1300 ms, inertial data (including acceleration and angular velocity) was recorded at a frequency of 100 Hz. In line with the data input approach of Zhou et al. [14], gait events were divided and linearly interpolated to form a fixed-length sequence of 64 sample points. The dataset was developed by combining the acceleration data and the angular velocity data.

## IV. Deep Learning-Based Disguised Inertial Gait Recognition Method

To address the challenge of disguised inertial gait recognition, we adopt a Convolutional Recurrent Neural Network (CRNN) as the baseline method. Following Wu [13] and Zhou [14], we design a recognition framework capable of extracting spatial and temporal gait features for recognition.

### A. Problem Formulation

Gait recognition can be formulated as an n-class classification problem, where a given test sample is identified among n candidate identities. Given an inertial gait signal $x$ of length $T$, it can be represented as:

$$x_t = (a_t, g_t), \quad t = 1, 2, \ldots, T \tag{1}$$

where $a_t = (acc_x^t, acc_y^t, acc_z^t)$ denotes the accelerometer on three axes at time $t$, and $g_t = (gyr_x^t, gyr_y^t, gyr_z^t)$ represents the gyroscope. The objective is to learn a classification function that maps input gait data $x$ to an identity class:


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$$p(s_i \mid x) = f(x),\ \ i = 1, 2, ..., n \tag{2}$$

where $p(s_i \mid x)$ denotes the probability of $x$ belonging to identity $s_i$.

### B. Network Structure

The proposed CRNN framework integrates CNNs and RNNs to model both spatial and temporal gait features, as shown in Figure 4. The CNN module extracts local spatial patterns from inertial data using three layers of 1D convolution, effectively reducing noise and individual variability while capturing multi-scale features.

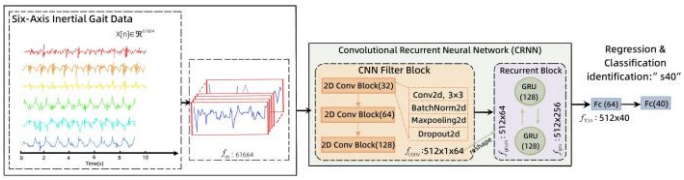


Fig. 4. Overview of the CRNN-based disguised inertial gait recognition method. The model extracts spatial and temporal gait features from the accelerometer and gyroscope, followed by classification through a fully connected layer.

The RNN module employs a bidirectional LSTM-GRU architecture, where LSTM captures long-term dependencies and GRU improves computational efficiency. The bidirectional design ensures the utilization of both past and future context. Finally, extracted features are passed through a fully connected layer and classified via a Softmax layer.

The model's feature transformation can be expressed as:

$$h_t = GRU(LSTM(CNN(X))) \tag{3}$$

$$y = softmax(Wh_t + b) \tag{4}$$

where $h_t$ is represents features extracted by CNN, LSTM, and GRU. W and b are the weights and biases of the FCs.

### C. Training Strategy

Cross-loss entropy is used as the loss function:

$$o_i' = \begin{cases} 1, & x \in s_i \\ 0, & other \end{cases} \tag{5}$$

$$\iota(o, o') = \sum_i^n o_i' \ln o_i + (1 - o_i') \ln(1 - o_i') \tag{6}$$

When $o_i \approx 0$, $o_i' = 0$, or $o_i \approx 1$, $o_i' = 1$, the cross-entropy is smaller. This property helps the network to converge.

We optimize the model using the AdamW optimizer, integrating a cosine annealing learning rate schedule, to accelerate convergence and prevent local minima. Additionally, data augmentation techniques, such as time masking and signal perturbation [17], are applied to enhance model robustness.

## V. Experiments and Results

### A. Dataset and Experimental Setup

After obtaining inertial gait events of 40 subjects, 4 divisions were set up using the 61,346 gait events, as shown in Table I, where each subject was acquired individually to exclude interferences [13, 18].

A detailed description of the collected gait events is summarized in Table I. The number of extracted gait events was 14408, 15490, 15820, and 15628, respectively.

Table I: Description of dataset partitions and collected gait events.

| Theme | Description of Program | Sample |
|---|---|---|
| I | Put on a coat, and shoe type one, walking anticlockwise. | 14,408 |
| II | Removed coat and walked clockwise, wearing shoe type one. | 15,490 |
| III | Walked anticlockwise, wearing shoe type two without the coat. | 15,820 |
| IV | Wore the uniform lab coat and walked clockwise, with shoe type two. | 15,628 |

### B. Baseline Results of Disguised Inertial Gait Recognition

The baseline recognition results are shown in Table II. "None" covariate indicates no covariate influence. All the gait events in themes I, II, III, and IV were randomly mixed and split into two parts: rd_1 and rd_2. For the clothing covariate, groups I and IV served as the training set, while groups II and III were the test set. When it came to shoe influence, groups I and II were the training set, and groups III and IV were the test set. Reverse experiments were also carried out to examine the priority of the influencing factors.

The CRNN model showed the highest accuracy across all scenarios, validating its effectiveness in dealing with disguise conditions. It was found that removing clothing led to an approximately 11% reduction in accuracy, and changing footwear caused a 15% decrease, which shows the footwear has a more profound impact on gait recognition.

### C. Identification Results Visualization

Figure 5 presents the recognition accuracy under different disguise conditions. As shown, most subjects achieve recognition accuracies above 90%, demonstrating the robustness of the proposed method across disguise variations. Recognition accuracies for subjects S06 and S07 are noticeably lower compared to others. A closer inspection of their inertial data revealed higher variability and irregular gait sequences, which likely led to reduced model confidence and classification accuracy. Such samples represent natural outliers within the dataset and do not affect the overall recognition performance trends observed in the DIGRD benchmark.


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Table II: Recognition accuracy (%) on the DIGRD dataset across different models and disguise conditions.

| Covariate | Setting | | SVM | | CNN | | CN+LS | | CRNN | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Trian | Test | Acc. | Mean | Acc. | Mean | Acc. | Mean | Acc. | Mean |
| None | rd_1 | rd_2 | 78.72 | 78.88 | 98.92 | 98.87 | 99.01 | 98.97 | 99.40 | 99.34 |
| | rd_2 | rd_1 | 79.04 | | 98.77 | | 98.92 | | 99.27 | |
| Clothing | I, IV | II, III | 61.65 | 60.78 | 86.19 | 85.55 | 87.65 | 87.25 | 89.04 | 88.78 |
| | II, III | I, IV | 59.90 | | 84.91 | | 86.85 | | 88.52 | |
| Shoe Type | I, II | III, IV | 56.93 | 55.33 | 83.41 | 81.98 | 85.26 | 83.68 | 86.06 | 84.85 |
| | III, IV | I, II | 53.72 | | 80.55 | | 82.10 | | 83.64 | |

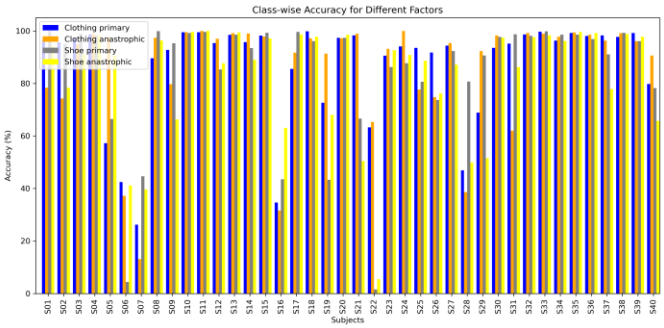


Fig. 5. Recognition accuracy on the DIGRD dataset containing both clothing and shoe type covariates. Clothing primary (sessions I, IV → II, III) and anastrophic, shoe primary (sessions I, II → III, IV), and anastrophic are shown in the figure with different colors.

### *D. Ablation Studies of Single Covariate-Based Recognition*

To further investigate the impact of single disguise factors, we conducted ablation studies on the DIGRD dataset, analyzing how each factor affects inertial gait recognition performance. Specifically, we structured the experiments as follows:

Clothing Variable: We evaluated the clothing impact by training on Theme I (thick coat) and testing on Theme II (no coat), and training on Theme IV (white lab coat) and testing on Theme III (no lab coat). This setup helps assess clothing's effect on recognition.

Footwear Variable: Reciprocal testing was used to assess footwear influence: train on Theme III (shoe type 2, no coat) and test on Theme II (shoe type 1, no coat), and vice versa. This evaluates the impact of shoe changes.

Combined Clothing and Footwear Variables: To mimic real-world scenarios, we conducted cross-variational tests. For example, train on Theme III (shoe type 2, no coat) and test on Theme I (shoe type 1, thick coat), etc. These tests show how combined factors affect recognition performance.

The results of these experiments are summarized in Table III. We can see that when considering individual disguise factors, their impact on recognition performance becomes more pronounced. Compared to the baseline accuracy of over 99% in the absence of disguise factors, clothing changes alone reduced accuracy to an average of 71.36% (thick coat) and 78.01% (white lab coat). Footwear changes had an even greater impact, with accuracy dropping by approximately 17%. In cases where both disguise factors were present, recognition accuracy further declined, demonstrating the combined effect of clothing and footwear disguises.

Table III: Recognition accuracy (%) for ablation experiments concerning the single covariate of clothing or shoe types.

| Covariate | Setting | | CRNN | |
|---|---|---|---|---|
| | Trian | Test | Acc. | Mean |
| None | rd_one | rd_two | 99.40 | 99.34 |
| | rd_two | rd_one | 99.27 | |
| Clothing | II | I | 71.79 | 71.36 |
| | I | II | 70.93 | |
| | IV | III | 77.08 | 78.01 |
| | III | IV | 78.93 | |
| Shoe Type | II | III | 73.38 | 72.84 |
| | III | II | 72.29 | |
| Cloth + Shoe | III | I | 71.65 | 72.85 |
| | I | III | 74.05 | |
| | IV | I | 65.73 | 67.01 |
| | I | IV | 68.28 | |
| | IV | II | 78.25 | 79.34 |
| | II | IV | 80.43 | |

These results confirm that both clothing and footwear impact inertial gait recognition, with footwear variations causing greater disruption. Footwear directly alters gait dynamics, while clothing mainly affects upper-body motion. These findings underscore the need for robust models capable of mitigating covariate effects and enhancing tolerance to such variations.

### *E. Impact of Different Training Data Organization Strategies on Recognition Performance*

Table IV: Dataset variants with different segmentation strategies.

| Dataset | Description | Samples | |
|---|---|---|---|
| | | Train | test |
| DIGRD_G5 | Samples are cut into 500-ms segments. | 35,529 | 36,374 |
| DIGRD_G6 | Samples are separated every 600ms. | 29,610 | 30,322 |
| DIGRD_G7 | Samples are generated with 700ms. | 25,389 | 26,001 |

To further evaluate the effect of training data segmentation, we applied different time windowing strategies to extend the DIGRD dataset. Given that individual gait events typically last between 500ms and 700ms, we created multiple dataset variants using different time window sizes for segmentation, as shown in Table IV.

To analyze the effect of training samples on recognition, we trained models using DIGRD_G5, DIGRD_G6, and


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DIGRD_G7 with different training set sizes (1,000 to 8,000). We tested the recognition performance under the clothing covariate (sessions I and IV for training, sessions II and III for testing). The results are illustrated in Figure 6.

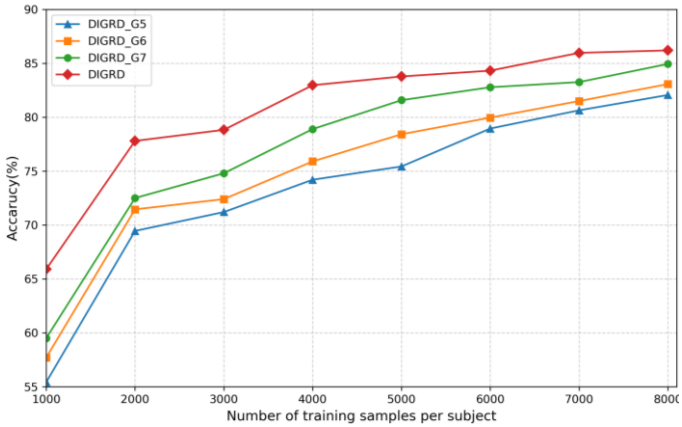


Fig. 6. Recognition accuracy distribution for DIGRD_G5, DIGRD_G6, DIGRD_G7, and the original DIGRD dataset under different training sample sizes.

The results indicate that recognition improves as the number of training samples increases. We also observe that datasets segmented with a 500-ms sliding window (DIGRD_G5) perform worse than those segmented with 600-ms or 700-ms. This suggests that excessively short segmentation may introduce unnecessary fragmentation, limiting the model's ability to capture full gait patterns. In contrast, segmentation strategies using 600ms and 700ms achieve higher accuracy, implying that moderate window sizes are more effective.

Moreover, when compared with the original DIGRD dataset (which uses individual gait event segmentation rather than fixed time windows), the DIGRD dataset achieves higher recognition accuracy. This highlights that segmenting complete gait cycles, rather than using fixed time windows, allows models to better focus on distinctive gait features, ultimately leading to improved recognition performance.

These findings provide insights into optimizing training data segmentation for inertial gait recognition, suggesting that appropriate time window selection is critical for maximizing recognition accuracy and robustness.

## VI. Conclusion

This paper presents DIGRD, the first benchmark dataset designed for disguised inertial gait recognition. Experimental results show that footwear changes reduce recognition accuracy to 84%, exerting a greater impact than clothing changes, which reduce accuracy to 87%. Compared with visual and acoustic methods, inertial sensing demonstrates stronger robustness but still exhibits vulnerability under disguise conditions.

Future work will focus on expanding the dataset with participants of diverse age groups, body types, and walking habits to enhance generalization. In addition, we plan to introduce stricter control over disguise variables, such as standardizing clothing materials, footwear types, and walking protocols, to minimize confounding effects and enable more precise evaluation of covariate influences.

## References


[1] A. Sepas-Moghaddam and A. Etemad, "Deep gait recognition: A survey," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 1, 2022.

[2] C. Shen, C. Fan, W. Wu, et al., "LidarGait: Benchmarking 3D gait recognition with point clouds," *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Vancouver, BC, Canada, 2023.

[3] W. Li, et al., "An in-depth exploration of person re-identification and gait recognition in cloth-changing conditions," *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.

[4] S. Zou, et al., "Cross-covariate gait recognition: A benchmark," *AAAI Conference on Artificial Intelligence*, Vol. 38. No. 7. 2024.

[5] M. Hofmann, S. Sural, and G. Rigoll, "Gait recognition in the presence of occlusion: A new dataset and baseline algorithms," *International Conference in Central Europe on Computer Graphics, Visualization and Computer Vision (WSCG)*, 2011.

[6] S. Zheng, J. Zhang, K. Huang, et al., "Robust view transformation model for gait recognition," *IEEE International Conference on Image Processing (ICIP)*, 2011.

[7] C. Xu, Y. Makihara, G. Ogi, et al., "The OU-ISIR gait database comprising the large population dataset with age and performance evaluation of age estimation," *IPSJ Transactions on Computer Vision & Applications*, vol. 9, 2017.

[8] S. Wu, X. Zhai, Z. Hu, et al., "Advanced acoustic footstep-based person identification dataset and method using multimodal feature fusion," *Knowledge-Based Systems*, vol. 264, 2023.

[9] S. Huang, R. Fan, and S. Wu, "DisGait: A prior work of gait recognition concerning disguised appearance and pose," *International Conference on Intelligent Computing (ICIC)*, 2023.

[10] C. Santos, D. Oliveira, L. Passos, et al., "Gait recognition based on deep learning: A survey," *Acm Computing Surveys (CSUR)*, vol. 55, no. 2, 2022.

[11] B. Lin, S. Zhang, and X. Yu, "Gait recognition via effective global-local feature representation and local temporal aggregation," *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

[12] L. Du, X. Chen, Y. Shi, et al., "MMRGait-1.0: A radar time-frequency spectrogram dataset for gait recognition under multi-view and multi-wearing conditions," *Journal of Radars*, vol. 12, no. 4, 2023.

[13] S. Wu, S. Huang, Z. Liu, et al., "AFPILD: Acoustic footstep dataset collected using one microphone array and LiDAR sensor for person identification and localization," *Information Fusion*, vol. 104, 2024.

[14] Q. Zou, Y. Wang, Q. Wang, et al., "Deep learning-based gait recognition using smartphones in the wild," *IEEE Transactions on Information Forensics and Security*, vol. 15, 2020.

[15] K. Ma, Y. Fu, D. Zheng, et al., "Dynamic aggregated network for gait recognition," *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.

[16] G. Prateek, P. Mazzoni, G. Earhart, and A. Nehorai, "Gait cycle validation and segmentation using inertial sensors," *IEEE Transactions on Biomedical Engineering*, vol. 67, no. 8, 2020.

[17] S. Wu, Y. Wang, Y. Jiang, et al., "CRATI: Contrastive representation-based multimodal sound event localization and detection," *Knowledge-Based Systems*, vol. 305, 2024.

[18] S. Wu, J. Guang, W. Wu, et al., "Dfsc-da: Dominant frequency segmented conformer with domain adaptation for acoustic footstep-based person identification," *Expert Systems with Applications*, vol. 294, 2025.




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